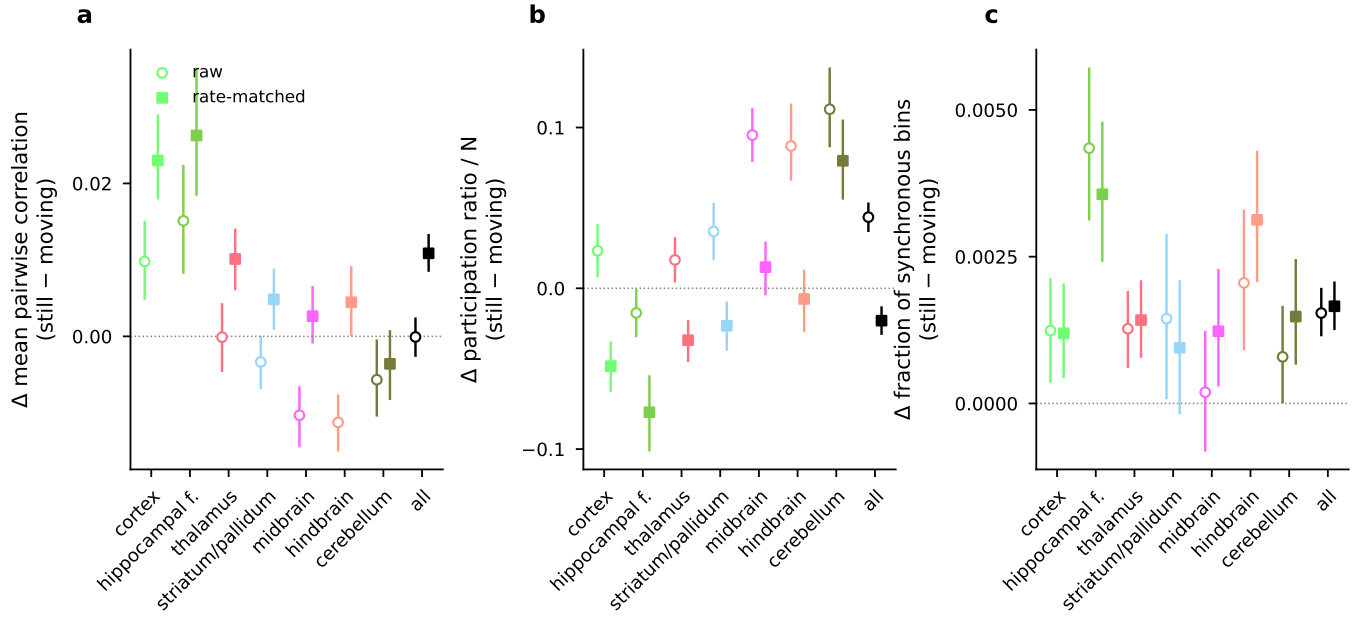

Figure 5: In stillness, neurons within a region fluctuate together more (rate-matched), most clearly in cortex and hippocampus. **a** Mean pairwise spike-count correlation within a region (100 ms bins, equal numbers of still and moving bins), still minus moving, per region and probe averaged within mouse; mean over mice and bootstrap 95% CI. Open: raw counts; filled: each neuron's counts in its higher-rate state binomially thinned to its rate in the other state. Rate-matched, 82% of 129 mice have higher correlation in stillness (cortex 91%). **b** Participation ratio of the correlation matrix divided by the number of neurons (dimensionality). The raw increase in stillness disappears after rate matching. **c** Fraction of 100 ms bins in which the summed population count exceeds its within-state mean by 3 SD (synchronous events).